



Progress Report

Student name: Mohamed Redha Alaraibi

Student ID: 202101014

Course Title: EN8914 - CL Project

Project Title: SmartGuard

Supervisor's Name: Dr. Zakareya Hasan

Table of Contents

Project overview	4
Project purpose	4
Report purpose	4
Introduction	5
Phase-1: Identifying the problem	6
Outcomes	8
Phase-2: Research and planning	8
Changes	10
Outcomes	11
Phase-3: Select the Components.....	13
Successes and Challenges	14
Changes	15
Outcomes	15
Phase-4: Software development.....	17
Camera system development	17
Sensor system development	19
Successes and Challenges	22
Changes	23
Outcomes	23
Next steps.....	25
Phase-5: Hardware development and Prototype	26
Objectives.....	26
Potential Challenges	27
Expected Outcomes	28
Next step work plan.....	29

Phase-6: Integration and Testing	30
Objectives.....	30
Potential Challenges	31
Expected Outcomes	32
Phase-7: Finalization, Report, and Presentation	32
Objectives.....	32
Potential Challenges	33
Expected Outcomes	33
Gantt Chart Modifications.....	34
Conclusion and Recommendations.....	36
Recommendations.....	36

Table of tables:

Table 1: The initial Bill of Materials (BOM)	16
Table 2: The finalized Bill of Materials (BOM)	25

Table of figures:

Figure 1: The initial Gantt Chart.....	12
Figure 2: Initial Gantt Chart's Details.....	12
Figure 3: System Diagram	13
Figure 4: The Modified Gantt Chart.....	35
Figure 5: Modified Gantt Chart's Details	35

Project overview

This project offers an intelligent monitoring system for children's transportation buses that combines artificial intelligence with a number of sensors and smart features in order to improve safety and stop suffocation accidents. Using a number of sensor modules, the system continuously determines whether children are inside the bus. Continuous location tracking is made possible by real-time GPS data, and all sensor data is processed by a central microcontroller before being sent over LTE-NB IoT technology. The technology immediately initiates a series of safety measures, such as instant alerts to supervisors and guardians, the activation of warning indicators, and automated escalation procedures, if a child is found inside the bus after it has been turned OFF. When combined, these elements provide a dependable, clever, and proactive way to guarantee children's safety throughout regular bus operations.

Project purpose

The main purpose of this project is to develop a reliable, intelligent monitoring system that enhances the safety of children in school and kindergarten buses. With recurring incidents of children being forgotten inside buses in Bahrain and the Gulf region, the project aims to create an automated solution capable of detecting a child's presence and environmental conditions in real time. By integrating advanced sensors, GPS tracking, and artificial intelligence, the system provides immediate alerts and automated safety actions, offering a practical and effective approach to preventing suffocation incidents and ensuring safer daily transportation.

Report purpose

The main purpose of this progress report is to present an updated overview of the development status of the intelligent child-safety monitoring system for school buses. It summarizes the work completed in each phase, identifies the challenges encountered, and outlines the solutions implemented and lessons learned. The report also highlights upcoming tasks, planned improvements, and the project timeline in the form of a Gantt chart. Overall, this report ensures that the project remains aligned with its objectives, schedule, and intended safety goals.

Introduction

In recent years, the Gulf region and Bahrain in particular have witnessed an alarming increase in tragic incidents involving children being forgotten inside school buses, often leading to severe heat-related complications and, in some cases, fatal outcomes. These events have created a sense of urgency within educational communities and families, highlighting the critical need for innovative, reliable, and sustainable solutions that can prevent such incidents and protect children's lives. Motivated by these real-world concerns and a shared responsibility toward student safety, our team set out to develop a comprehensive monitoring system that applies advanced technologies to address this issue effectively.

This project focuses on developing an integrated monitoring system for children's transportation buses, designed to enhance student safety and provide dependable real-time information to guardians and school administrators. Given the increasing risks associated with unattended children in buses and unsafe boarding conditions, there is a pressing need for an intelligent and automated solution capable of monitoring occupancy, environmental parameters, and bus status with high precision. Our system aims to ensure continuous supervision, minimize human error, and support swift decision-making during daily bus operations.

The proposed solution combines artificial intelligence techniques, a suite of advanced sensors, and the practical knowledge and technical skills we acquired throughout our engineering studies.

Project phases

The development of our project has been organized into a series of clear and sequential phases that allowed us to maintain structured progress and monitor our outcomes at every stage. From the very beginning, we prepared a detailed work plan to guide our implementation within the available timeframe, helping us track milestones, allocate tasks effectively, and adapt whenever new requirements or challenges appeared. Like any real engineering project, our journey has not been perfectly linear; each phase introduced its own technical challenges and unexpected complications. These situations often required us to revisit earlier decisions, adjust our system design, or modify our implementation strategies in order to reach practical and reliable solutions.

As we progressed, several major components and methodologies were revised through an iterative problem-solving approach. These adjustments were essential for overcoming critical issues, improving system performance, and ensuring that our final solution remains aligned with the project's objectives. In this progress report, we present a detailed overview of each phase we have completed so far, the challenges we faced, the changes we applied, and the reasoning behind the decisions we made. By documenting the development process step by step, the report reflects not only our technical achievements but also the continuous learning and refinement that shaped the project into its current form. Additionally, this report outlines the upcoming tasks we plan to work on in the next weeks, providing a clear roadmap for the remaining development stages.

Phase-1: Identifying the problem

The project originated from our collective commitment to addressing a critical issue that continues to impact communities across the Gulf region and Bahrain in particular: the recurring incidents of children being left unattended inside kindergarten and school buses, leading to severe health risks and, in some cases, tragic loss of life. Motivated by the urgency of this challenge, we began our idea-generation phase through a structured and organized approach. Each team member initiated individual research to explore innovative concepts that could realistically be implemented within the scope of a senior engineering project.

During this initial stage, we also leveraged our internship experiences by consulting professional engineers, supervisors, and field experts to gather technical insights, practical advice, and potential solution ideas. Additionally, we engaged with several university professors to benefit from their academic perspectives and guidance. These interactions broadened our understanding, improved the practicality of our ideas, and encouraged us to incorporate modern technological methods.

After completing the information-gathering phase, we conducted a series of focused brainstorming sessions where we presented, discussed, and critically evaluated all proposed ideas. These meetings were highly collaborative: each member presented their ideas in detail, and together we assessed feasibility, innovation level, societal impact, and technical requirements. Our discussions were supported by additional research to validate assumptions and identify potential implementation challenges.

Through this comprehensive evaluation, the idea that emerged as the most impactful and technically promising was the development of an intelligent child-safety and monitoring system for school buses, a system designed to prevent suffocation incidents caused by children being forgotten inside a bus after shutdown. This concept aligned strongly with regional needs, matched global trends in smart safety systems, and demonstrated high potential for real-world application.

The formal start of our project took place in mid-September, when we reached a consensus on the core concept: a system capable of continuously monitoring the interior of a bus using a sensor and a camera, detecting the presence of a child, and issuing immediate alerts. With the main idea clearly defined, we presented the proposal to our project supervisor, who expressed strong support and approved the concept, recognizing its relevance, practicality, and expected impact. This approval marked the official beginning of the development phase and served as a significant motivation for our team.

Our initial concept relied on two components: a millimeter-wave radar sensor capable of detecting human presence and tracking subtle movements, and a live-streaming camera that monitors any person inside the bus. The system's basic procedure involved sending simple alerts to bus drivers if someone was detected inside the bus after it had been turned off.

Outcomes

The problem identification phase provided us with a clear understanding of the critical safety issue facing children in school and kindergarten buses within Bahrain and the Gulf region.

Through our research, consultations, and brainstorming sessions, we successfully defined the main objective of our project: to develop an intelligent system capable of detecting the presence of children inside buses after shutdown and sending immediate alerts to prevent suffocation incidents. This phase also allowed us to establish our initial system concept, which integrates a millimeter-wave radar sensor and a live-streaming camera as the primary detection tools.

Overall, the outcomes of this stage provided us with a strong foundation, a well-defined problem scope, and a practical direction for the upcoming design and implementation stages.

Phase-2: Research and planning

After our project idea was approved, our next step was to explore ways to build it, develop it further, and expand its capabilities by adding features that would increase its value and effectiveness, ultimately helping us create a sustainable solution to a problem our community has long faced.

Once the concept received approval, we moved into the research and planning stage. We conducted several meetings, both in person and online. At that time, our core idea was based on using a single camera and a C1001 mmWave human-detection sensor to identify the presence of anyone inside the bus, with alerts or messages sent whenever detection occurred. During these discussions, each of us shared different scenarios, ideas, features, and potential innovations. It quickly became clear that the idea required deeper and more detailed study to reach an optimal solution capable of addressing the recurring problem of children being forgotten inside buses after the engine is turned off.

As a team, we recognized the need to add additional features during the development phase to enhance the system's efficiency and further reduce the chances of anyone being left behind. This research and planning stage marked the beginning of our project's design and implementation strategy. It involved expanding the project and introducing several new features after continuous consultations among team members and with our supervisor. This phase essentially served as our

roadmap, guiding the refinement of the initial idea and laying out the many added components that eventually made the project more advanced and comprehensive.

Throughout this stage, we relied heavily on the diverse perspectives within our team. The variety of viewpoints and backgrounds played a crucial role in shaping the project's success. Each member approached the idea from a different angle, contributing suggestions and enhancements that transformed the concept into a complete, practical, and sustainable solution. After numerous meetings, our goal remained clear: to build upon the idea defined in the previous phase.

Although the original concept was strong, it had shortcomings in several important areas, making it necessary for us to implement multiple changes and additions to increase its reliability and overall effectiveness.

Through the meetings we held among the team and with our supervisor, as well as consultations with experienced individuals, several additions were adopted to complement the basic components: the millimeter-wave human detection radar sensor and the live-streaming human detection camera. The adopted additions are as follows:

1. Add an environmental sensor to display the ambience conditions inside the buses (temperature and humidity).
2. Add oxygen sensor to display the oxygen percentage inside the buses.
3. Add IR sensors at the bus entrances. These sensors act as a counter to count the number of children inside the bus.
4. Add a load sensor to detect the presence of people sitting in the seats.
5. Add a display screen for the driver showing the number of passengers, temperature, and humidity.
6. Add a GPS device to determine the bus's location.
7. Create a mobile application or dashboard that allows parents and school/kindergarten administrators to monitor the bus, and displays: the number of passengers, temperature and humidity, oxygen levels, bus location, and a live stream showing the passengers inside the bus.

8. Adding a series of automatic actions that occur if a child is detected inside the bus after it has been turned off. These actions are as follows:

- Opening the bus window
- Opening the bus door
- Flashing the bus headlights
- Sounding the horn
- Sending alerts to drivers and all application or Dashboard users

After we completed the modifications and additions to our project idea, our next step was to prepare the essential project documents. We divided the work among ourselves: two team members focused on developing the project proposal, which included detailing the idea, objectives, key features, and overall concept; another member created the Gantt chart to illustrate the project timeline and how the tasks were distributed; while the fourth member designed an initial diagram showing the main project components, along with a preliminary layout for the application or dashboard.

Changes

During the research and planning phase, our project idea evolved through several major enhancements that we introduced to improve its functionality, reliability, and long-term sustainability. What initially began as a simple system built around a single camera and a C1001 mmWave human-detection sensor gradually expanded into a more comprehensive solution as we continued our analysis and discussions.

As a team, we decided to integrate a wider range of sensors and intelligent features, including environmental monitoring sensors for temperature, humidity, and oxygen levels, infrared sensors for passenger counting, a load sensor to detect whether someone is seated, and a GPS module for real-time location tracking. We also added a driver display and a mobile application or dashboard to present all system data such as live video, passenger count, and environmental conditions in a clear and accessible way.

Finally, we incorporated an automated emergency response mechanism that allows the bus to take predefined safety actions when a child is detected inside after shutdown. These actions include opening doors and windows, flashing the external lights, sounding the horn, and sending instant alerts. Through these additions, the project transitioned from a basic detection concept into a fully integrated, intelligent safety and monitoring system—far more capable, robust, and practical than the original idea.

Outcomes

The research and planning phase provided us with a clear and well-organized roadmap that guided our transition from the initial concept into actual implementation steps. Throughout this stage, our project grew into a fully developed smart system that combines artificial intelligence, integrated sensors, and automated safety mechanisms. With continuous teamwork, valuable input from our supervisor, and consultations with experienced professionals, we were able to refine the original idea into a solution that is both feasible and scalable while also remaining innovative.

During this phase, we also completed the essential project documents, including the detailed project proposal, the Gantt chart that outlined the project timeline and task distribution, and preliminary diagrams illustrating the system's components along with the initial layout of the mobile application/dashboard.

Overall, this phase established a strong foundation for the upcoming design and development stages, ensuring that our work remains aligned with the project's core objective: developing a reliable, intelligent, and sustainable system to prevent child suffocation incidents inside school and kindergarten buses.

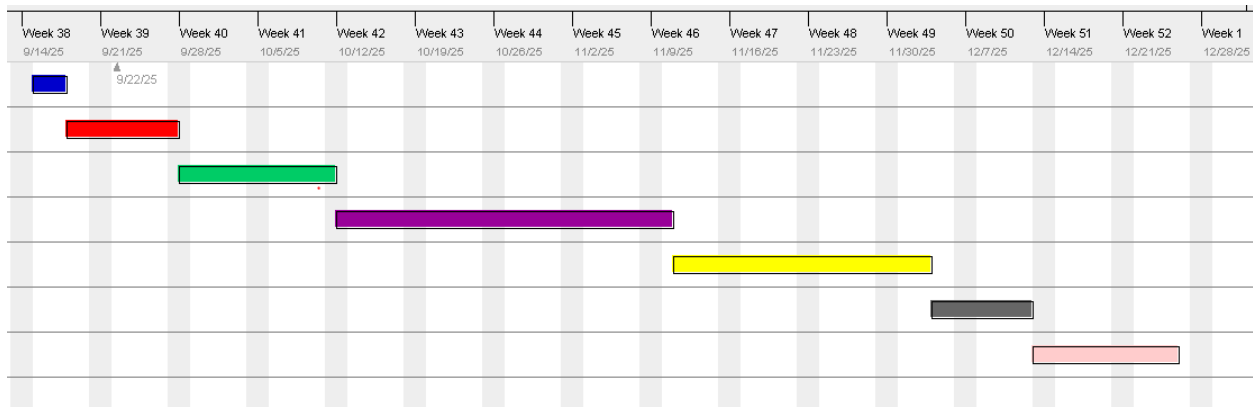


Figure 1: The initial Gantt Chart

Name		Begin date	End date	Duration
Identifying The Problem	●	9/15/25	9/17/25	3
Research and Planning	●	9/18/25	9/27/25	10
Select the Components	●	9/28/25	10/11/25	14
Software Development	●	10/12/25	11/10/25	30
Hardware Development	●	11/11/25	12/3/25	23
Integration and Testing	●	12/4/25	12/12/25	9
Finalization, Report, and Presentation	●	12/13/25	12/25/25	13

Figure 2: Initial Gantt Chart's Details

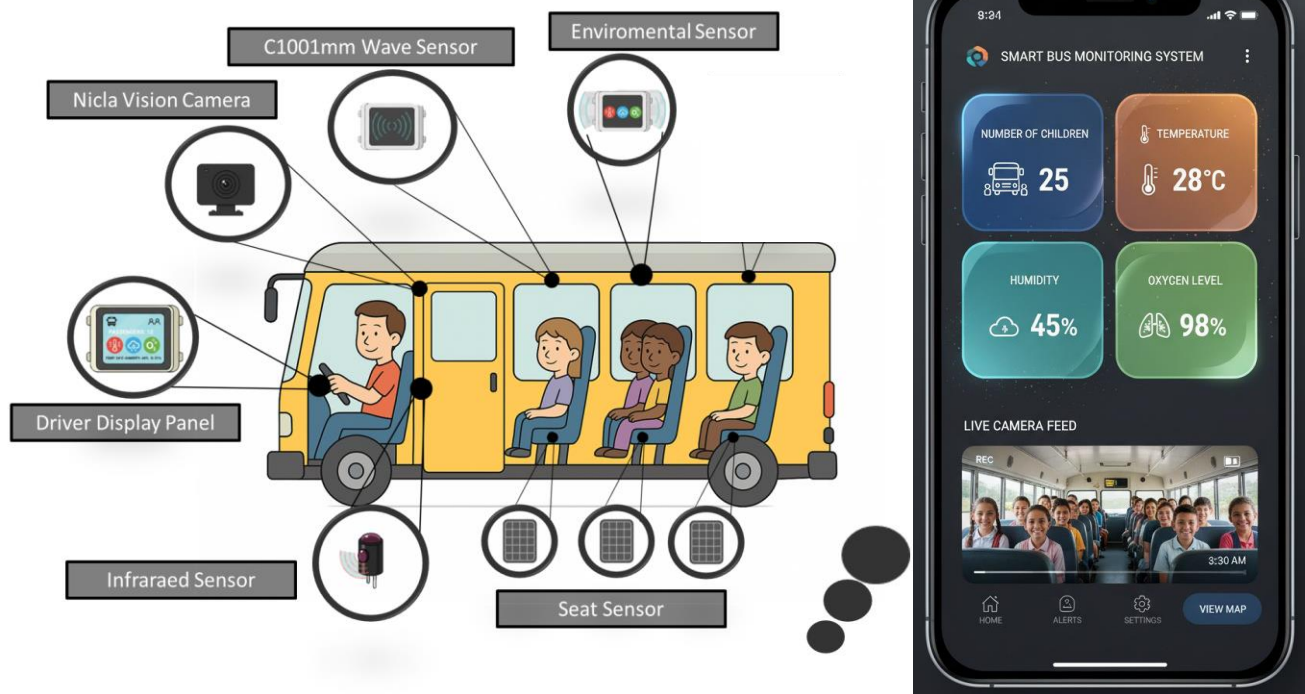


Figure 3: System Diagram

Phase-3: Select the Components

Based on the documents we produced in the previous phase, including the project proposal, the Gantt chart, and the system diagrams, we moved forward into the component selection stage. During this phase, we held a series of team meetings to finalize the features and functionalities that would be included in the system. These discussions helped us determine the requirements for the main microcontroller, which serves as the core of the entire system by reading sensor inputs, processing data, and executing automated responses.

We evaluated several microcontroller options, considering performance, cost, compatibility, and available features. After extensive research and collaborative discussions, we agreed on selecting the Arduino MKR NB 1500. Its strong capabilities, reasonable cost, and built-in SIM card slot which enables direct transmission of alerts and messages, made it the most suitable choice for meeting the project's safety requirements.

To streamline the work, we divided the team into two groups. The first group focused on researching camera options, comparing prices, specifications, and programming compatibility. Most commercial cameras required Python programming and Raspberry Pi, but during the investigation we came across the Arduino Nicla Vision camera, designed specifically for Arduino platforms. This eliminated the need for an additional microcontroller. Although information was initially limited, further testing and demonstrations confirmed that the camera was capable of both human detection and live streaming, making it a promising option for our system.

The second group worked on selecting the sensors required for the project's various functions. For each feature, we identified multiple sensor options, evaluated their performance, reliability, advantages, disadvantages, and cost, and then selected the most suitable one to ensure high accuracy and overall system efficiency.

After finalizing all components, we compiled a complete Bill of Materials (BOM) that included the specifications and purchasing links for every item. We also conducted additional research to find reliable suppliers offering the best prices. Once this was completed, the components were ordered, marking the transition from planning to the practical implementation stage of our project.

Successes and Challenges

During this phase, our component selection process progressed smoothly and according to plan. As a team, we were able to conduct productive meetings to finalize the system's features, research the available components, and choose the most suitable options based on performance, reliability, and cost. We also completed the preparation of the Bill of Materials (BOM) and successfully identified appropriate suppliers, allowing us to finalize the ordering process efficiently.

However, once the component orders were placed, we encountered several unexpected challenges. Some key components, specifically the Arduino MKR NB 1500 and the Arduino Nicla Vision camera experienced delivery delays due to shipping issues. In addition, the GPS module arrived damaged, which required us to contact the supplier and request a replacement.

Although the supplier responded positively, the replacement procedure introduced additional delays.

These issues affected our implementation schedule and required us to adjust the upcoming timeline, including making updates to the Gantt chart. Despite these setbacks, we managed the situation effectively through team coordination, continuous communication, and proactive problem-solving, ensuring that the project remained on track.

Changes

To overcome the challenges, we faced during the component delivery stage, we implemented several adjustments. We reviewed and updated the project timeline based on the revised delivery dates to ensure that the delays would not significantly affect our overall progress. Accordingly, we modified the Gantt chart to maintain realistic milestones and deadlines that aligned with the new situation.

During the waiting period for the delayed and replacement components, we made effective use of the available time by preparing for the upcoming implementation stage. This included reviewing sensor interfacing methods, studying the data communication protocols required for the Arduino MKR NB 1500, and finalizing our initial testing plan. These proactive steps helped reduce the impact of the delays and allowed us to keep the project moving steadily toward successful implementation.

Outcomes

By the end of this phase, all selected components were successfully received in good condition, completing the procurement process. The finalized list of components reflects careful consideration of functionality, compatibility, reliability, and cost-effectiveness. The table below summarizes the selected components that form the foundation of the smart monitoring system.

Component	Description / Model	Selection Criteria
Microcontroller	Arduino MKR NB 1500	Integrated LTE-NB IoT support, SIM slot for direct messaging, reliable performance
Human Detection Sensor	C1001 mmWave Human Detector	Accurate human presence detection, high sensitivity, reliable range
Camera Module	Arduino Nicla Vision	AI capability for human detection and live streaming, Arduino compatibility
Temperature & Humidity Sensor	DHT22	High accuracy, stable reading, widely supported
Oxygen Sensor	US10010-1	Oxygen level measurement, easy integration
Load Sensor	HX711	Precise weight detection, suitable for occupancy sensing
IR Sensors	MH-Sensor-Series-Flying-Fish	Accurate entry/exit counting, quick response
Display Screen	LCD	Real-time display for driver interface
GPS Module	MKR GPS Shield	Compatible with Arduino MKR board, accurate real-time location tracking

Table 1: The initial Bill of Materials (BOM)

Despite the delays encountered, this stage was ultimately successful. All components were procured, verified, and documented, providing a strong foundation for the next stage of the project. This phase represents the beginning of the practical implementation of the system, where the hardware integration and programming will commence to bring the designed concept to life.

Phase-4: Software development

After we successfully received and verified all hardware components, we moved on to the software development phase, marking our transition from planning to actual system implementation. In this stage, we focused on programming the microcontroller and camera modules, integrating the sensors, and developing the communication and monitoring interface. Our main objective was to ensure smooth interaction between hardware and software components, enabling accurate data collection, real-time monitoring, and automated safety responses as outlined in our project design.

We made efficient use of the waiting period during component delivery by conducting in-depth research on the functionality and programming requirements of each module. This proactive approach allowed us to transition smoothly into hands-on implementation once all components arrived. After confirming that all parts were intact and functional, we divided our team into two specialized groups: one group focused on the camera system, while the other concentrated on the sensors system.

Camera system development

During the waiting period for component delivery, the camera team dedicated their efforts to researching the Arduino Nicla Vision camera, studying its technical specifications, programming methods, and potential integration within the project system. Once all components arrived and their integrity was verified, the team divided responsibilities, one member focused on developing the human detection feature, while the other worked on implementing the live streaming functionality.

The Arduino Nicla Vision camera was initially selected due to its compact size, direct compatibility with Arduino hardware, and built-in support for basic image processing. However, practical implementation proved far more challenging than anticipated. The available documentation and resources for the camera were limited, and very few open-source examples existed for the required tasks. The team began testing using the Arduino IDE, but all early attempts to execute human detection algorithms failed due to insufficient library support and memory limitations.

After extensive troubleshooting, the team decided to shift to the OpenMV environment, which offered more flexibility for visual processing and object recognition. This transition significantly improved development progress. After several days of trials and code adjustments, the human detection feature was successfully implemented. The system was able to identify human presence within a defined area, but limitations soon became apparent: the detection range was short, and the accuracy decreased under low-light or variable lighting conditions. Similarly, the live streaming function was implemented after several iterations, but the video quality remained moderate, and the data transmission rate was inconsistent.

Recognizing that these issues would negatively affect system performance and reliability, the entire team convened along with our supervisor, to evaluate potential solutions. After detailed discussions, we decided to replace the Arduino Nicla Vision camera with a more advanced vision system based on the Raspberry Pi platform. This change introduced greater processing capability and support for powerful Python-based computer vision libraries such as OpenCV and MediaPipe, enabling more sophisticated algorithms and higher-quality image handling.

Under the new setup, two Raspberry Pi-compatible cameras were introduced (They were borrowed from a friend to save time and prevent any delays in the project), each assigned a specific function to maximize efficiency.

The first camera was tasked with performing human detection and live streaming inside the bus, continuously monitoring for any passengers left behind after trips.

The second camera was designed to implement face recognition and automatic student counting, a feature that significantly elevated the project's innovation and functional value.

The same two team members continued to handle this subsystem, one focusing on the first camera's dual functionality and the other specializing in developing the face recognition feature. The first camera was successfully programmed and tested, performing both real-time human detection and live video streaming reliably after multiple optimization stages. Network latency and frame rate were improved through parameter tuning, achieving a stable live feed suitable for real-time monitoring.

For the face recognition camera, the development process is ongoing but has achieved notable progress. To build and train the recognition model, the team captured a dataset of facial images

from team members and university instructors, allowing for local testing and verification. The model successfully identified known faces during early testing, and current efforts focus on improving detection accuracy and processing speed.

This transition from the Arduino Nicla Vision to the Raspberry Pi-based system marked a major advancement in both functionality and technical robustness. The new configuration not only resolved the earlier limitations of range and image quality but also added an innovative facial recognition feature that strengthens the project's goal of ensuring maximum safety and accountability for children inside buses.

The success of this phase demonstrated the team's ability to adapt to technical challenges through systematic problem-solving, research, and collaboration, ultimately leading to a more intelligent, scalable, and dependable vision subsystem.

Sensor system development

The sensor development group began working immediately after component delivery by individually connecting, programming, and testing each sensor on the Arduino MKR NB 1500 before proceeding to full system integration. This process allowed the team to identify and solve hardware and software issues early, ensuring reliable performance in the final setup.

The temperature and humidity sensor (DHT22) was the first to be implemented. Its configuration and programming were straightforward, and the sensor responded with accurate, stable, and consistent readings under different environmental conditions. This confirmed its suitability for continuous in-bus environmental monitoring.

Next, the team worked on the load sensor (HX711), responsible for detecting the presence of passengers based on seat pressure. This sensor required an initial calibration phase to ensure accuracy, but early attempts produced fluctuating and incorrect readings due to instability in sensor placement and uneven pressure distribution. After thorough troubleshooting and research, the team discovered that the calibration errors were caused by inconsistent force application points. To solve this, custom 3D-printed mounting bases were designed and fabricated, providing a stable and uniform surface for accurate load measurements. Once implemented, the readings became precise, and the calibration process stabilized successfully.

The IR sensors were then programmed and tested for the passenger counting mechanism at bus doors. The initial pair of sensors exhibited limited performance their medium detection range and slow response time led to inaccurate counting, especially when students entered or exited quickly or in groups. The team ran several tests, rewrote portions of the detection code, and modified sensor alignment, but results remained unsatisfactory. To address this, an intensive search for better alternatives available locally in Bahrain was conducted. The team identified the E18-D80NK sensors, which feature adjustable range, fast response, and higher accuracy. These were purchased, installed, and programmed with a dual-beam logic configuration: when a person breaks the first beam before the second, the system registers an “entry”; when the sequence is reversed, it registers an “exit.” This logic proved highly reliable, as confirmed by multiple tests showing accurate and real-time passenger counting with no false readings.

The oxygen sensor (US10010-1) was also tested as part of the system’s safety and environmental monitoring features. However, during testing, the team discovered that the sensor could not provide real-time and direct oxygen level measurements as required for this application. After additional research and analysis, the team found that suitable replacement sensors were significantly more expensive and would exceed the project’s budget. After a detailed discussion among team members, it was decided to remove the oxygen-sensing feature from the current version of the project to maintain cost-effectiveness and ensure focus on the core functionalities.

The C1001 mmWave human detection sensor posed one of the greatest technical challenges in this phase. The sensor required highly controlled calibration conditions, including an empty, motion-free environment. Even minor movements during setup caused calibration failure or inaccurate readings. Furthermore, the sensor’s performance was highly sensitive to installation orientation and distance from reflective surfaces. After several unsuccessful calibration attempts, the team conducted deeper research to understand the sensor’s internal processing algorithm and timing requirements. Over several days of careful adjustments in both hardware positioning and code structure, the team gradually improved the sensor’s reliability until stable and accurate presence detection was achieved. The experience provided valuable insight into mmWave sensing principles and improved the overall system design.

To further enhance the reliability and accuracy of the system’s human detection feature, the team decided to integrate an additional thermal imaging sensor. This sensor detects human presence

based on body heat signatures, allowing for an independent verification of the mmWave sensor's data. After researching available models, an appropriate sensor was procured, which is (AMG8833), programmed, and tested. The results showed excellent accuracy and sensitivity, even under varying temperature and lighting conditions. The inclusion of this sensor not only increased the system's redundancy and dependability but also added an innovative dimension to the monitoring system, setting it apart from conventional solutions.

After completing individual testing, all sensors were integrated into a unified system. During the early integration trials, several interference and signal conflict issues arose between sensors, leading to inconsistent readings. These issues were traced to timing conflicts and overlapping I/O operations within the Arduino code. Through detailed debugging, timing adjustments, and communication protocol optimization, the team successfully synchronized all sensors, achieving stable and coordinated operation across the full system.

To simulate automatic emergency response procedures, the team used LED indicators as temporary actuators representing real components (alarms, window openers, or horn). During this stage, a hardware limitation was discovered the Arduino MKR NB 1500 did not have enough I/O pins to connect all components simultaneously. After research, the team identified a suitable expansion module, the PCF8575 I/O expander, which communicates via the I2C protocol. This module was installed and successfully increased the available pin count, allowing all devices to operate concurrently. Additionally, a toggle switch was added to represent the bus's power state (ON/OFF).

Finally, the control logic was implemented to ensure accurate and safe operation of the system. The programmed conditions required that the bus must be turned off and at least two different sensors must simultaneously detect human presence before triggering emergency alerts or safety actions. This multi-sensor validation mechanism significantly reduced false alarms and enhanced the overall system reliability and credibility. After multiple tests, the complete sensor and control system demonstrated accurate, consistent, and dependable performance.

Currently, the team is working on integrating the GPS module (MKR GPS Shield) to provide real-time bus location tracking. While the GPS module functions correctly in standalone tests, a hardware conflict was discovered during full integration both the GPS and the C1001 mmWave sensor use the same Tx/Rx serial communication pins, which cannot be shared simultaneously.

The team is actively researching potential solutions, including hardware multiplexing or communication reassignment, to resolve this issue without delaying the project timeline.

Successes and Challenges

During the software development phase, the team successfully progressed from individual component testing to full system integration. Two specialized sub-teams were formed, one focusing on the camera subsystem and the other on the sensor subsystem to accelerate development and improve efficiency.

The camera team initially worked with the Arduino Nicla Vision, attempting to program features for human detection and live streaming. Despite extensive research and coding trials in the Arduino IDE and later OpenMV, the system suffered from limited range, poor image quality, and restricted processing capability. However, this phase was valuable for building a deeper understanding of embedded vision systems. Eventually, the team resolved these challenges by transitioning to a Raspberry Pi-based system equipped with two cameras: one for human detection and live streaming, and another for facial recognition and automatic student counting. This upgrade greatly enhanced system functionality and reliability.

Meanwhile, the sensor team successfully programmed and tested multiple sensors:

- The DHT22 functioned with excellent accuracy.
- The HX711 load sensor required careful calibration, solved by designing custom 3D-printed bases.
- The IR sensors were replaced with more advanced E18-D80NK models after repeated inaccuracies, leading to precise student entry and exit detection.
- The oxygen sensor was removed after proving unsuitable for real-time monitoring.
- The C1001 mmWave sensor posed calibration difficulties but was ultimately stabilized after several days of testing and code refinement.
- A thermal sensor (AMG8833) was added to enhance detection reliability through body-heat measurement.

Integration of all sensors revealed software conflicts that were later resolved through code optimization, ensuring stable and accurate readings across all modules. The team also

incorporated a PCF8575 I/O expander to overcome Arduino pin limitations and implemented logic conditions using LED indicators to simulate emergency actions marking a significant milestone in system testing.

Overall, the development process was productive and innovative. While several technical and hardware issues occurred, each challenge contributed to stronger teamwork, deeper understanding, and a more robust final design.

Changes

Throughout this stage, several important modifications were implemented to improve system performance and reliability:

- **Camera System Upgrade:** Due to technical limitations in the Arduino Nicla Vision, the team transitioned to a Raspberry Pi setup with dual cameras: one for live streaming and human detection, and another for facial recognition and automated student counting.
- **Sensor Replacement:** The IR sensors were replaced with E18-D80NK sensors, offering faster response and higher accuracy for entry and exit detection.
- **Feature Removal:** The oxygen sensor (US10010-1) was excluded after it failed to provide accurate, real-time oxygen readings, and viable alternatives were not cost-effective.
- **Hardware Enhancement:** A PCF8575 I/O expander was added to increase the number of digital pins available for component connections.
- **Additional Feature:** A thermal sensor (AMG8833) was integrated to enhance the detection of human presence by sensing body heat, improving overall system reliability.
- **Change in Actions:** We discovered that the busses windows do not operate automatically (they operate manually), which led us to dispense with the procedure of opening the windows.

These changes collectively optimized the system's design, enhanced functionality, and ensured smooth transition to the next implementation phase without major schedule disruptions.

Outcomes

By the end of the software development phase, the system achieved several measurable results reflecting technical progress and improved functionality. The camera subsystem successfully

performed real-time human detection and live streaming, while the facial recognition feature demonstrated promising results in identifying known faces during tests. The sensor subsystem achieved stable integration, accurate data readings, and consistent performance after debugging and calibration.

The following table summarizes the finalized list of components selected for the system, representing the Bill of Materials (BOM) achieved during this phase:

Component	Function	Remarks / Key Features
C1001 mmWave Human Detector	Detects motion and presence inside the bus	Calibrated successfully after multiple trials; high sensitivity
Arduino MKR NB 1500	Main microcontroller for sensor integration	Built-in SIM support for message alerts
Raspberry Pi with Dual Cameras	Human detection, live streaming, and facial recognition	Replaced Nicla Vision for enhanced performance
Temperature & Humidity Sensor (DHT22)	Measures bus interior temperature and humidity	Accurate and stable results
Load Sensor (HX711)	Detects seat occupancy or weight presence	Calibrated using custom 3D-printed bases
IR Sensors (E18-D80NK)	Counts students entering/exiting the bus	High response speed and precision
Thermal Sensor (AMG8833)	Detects human presence via body heat	Added to enhance reliability
LCD Display	Displays key system information	Functioning correctly with Arduino interface

GPS Module (MKR GPS Shield)	Tracks bus location in real-time	Successfully tested; integration ongoing
PCF8575 I/O Expander	Provides additional input/output pins	Solved port limitation issue

Table 2: The finalized Bill of Materials (BOM)

Next steps

In the upcoming stage, the main technical challenge to be addressed involves the UART communication conflict between the GPS module and the C1001 mmWave human detection sensor. Both components require the same serial communication pins (Tx and Rx) on the Arduino MKR NB 1500, which cannot be shared simultaneously due to hardware limitations. This overlap prevents both modules from transmitting and receiving data independently at the same time, making it necessary to establish an alternative communication method.

To resolve this issue, several potential solutions will be investigated and tested. One proposed approach is to integrate the SC16IS750 interface module, which operates through the I2C protocol and provides an additional UART interface. This would effectively expand the available serial communication channels, allowing both the GPS and the mmWave sensor to operate concurrently without interference. Another possible approach is to connect the GPS module via I²C using a specialized I2C-to-UART conversion cable, which would bypass the shared UART limitation entirely. The team will analyze both options in terms of reliability, data transmission stability, latency, and hardware complexity to determine the most suitable configuration.

After finalizing the communication setup, the GPS module will be fully integrated with the sensor network, and system-level testing will be conducted to ensure accurate location tracking and synchronization with the mmWave detection system.

In parallel, the development of the second camera system will continue. This camera will implement facial recognition to count the number of children accurately. Upon successful testing, the Raspberry Pi-based vision system will be linked with the Arduino control unit and sensors, completing the integration between sensing, vision, and control layers. LED lights will be used to simulate the system's automated response actions.

Furthermore, the team will configure and activate the SIM card module on the Arduino MKR NB 1500 to enable SMS alert functionality, a critical feature for real-time notification and safety assurance.

Once all components and communication channels are verified, the next milestone will focus on developing a centralized dashboard or mobile application. This interface will display real-time data, receive alerts, and serve as an accessible monitoring tool for parents, drivers, and school administrators, marking the transition to the fully functional and user-ready phase of the system.

Phase-5: Hardware development and Prototype

The hardware development stage represents the transition from software implementation to building a fully functional physical system. This stage focuses on assembling, integrating, and mechanically implementing all electronic components to create a realistic prototype that simulates an actual school bus environment. It includes designing and fabricating a custom PCB to organize the components into a compact, reliable unit, constructing a protective hardware enclosure, and building an interactive bus prototype that allows real-world testing of sensors, cameras, and emergency procedures. This is a future phase that will be carried out in the coming days and is essential for validating the system's readiness for real deployment.

Objectives

The hardware and prototype development stage aims to achieve the following objectives:

- Design and fabricate a PCB that integrates the Arduino, Raspberry Pi, cameras, and sensors into one clean, organized board.
- Construct a hardware enclosure (component box) to protect and neatly mount all electronics.
- Develop a realistic, interactive bus prototype that allows users to enter, sit, and trigger the system for testing.

- Implement a functional bus door mechanism that can open and close automatically to simulate real operational scenarios.
- Install lights, a horn, and motor to control the bus door to simulate emergency actions.
- Validate the complete system in a physical environment that closely represents school bus conditions.

Potential Challenges

Bus Door Design and Material Selection

One of the major challenges of this stage is designing a functional, realistic bus door. The team must determine appropriate materials that are:

- lightweight to reduce load on the motor,
- durable enough for repeated motion, and
- easy to mount on the wooden prototype.
- Materials such as lightweight MDF wood or acrylic panels will likely be used due to their strength and ease of fabrication.

Selecting the Motion Mechanism and Motor Type

A critical technical challenge is determining the most suitable motion mechanism for opening and closing the door. The team will evaluate multiple options, including:

- **High-torque Servo Motor:** suitable for rotational door movement with precise angle control.
- **DC Motor with Gearbox:** which provides higher torque for sliding or swinging doors but requires a motor driver.
- **Linear Actuator:** offers smooth linear motion like real bus doors and provides strong force for movement.

The final choice will depend on door weight, required motion type, and available mounting space.

Motor Control and Power Handling

Proper motor control is essential to ensure stable and safe operation. This includes:

- Using an H-bridge or motor driver for DC motors.

- Using PWM control in case of servo motors,
- Ensuring appropriate current supply,
- Preventing overload conditions, and
- Providing mechanical stability to avoid excessive vibration.

Ensuring Smooth Operation and Durability

We must ensure:

- Smooth opening motion without jerking.
- Correct door alignment.
- Stable mounting of the motor and hinges.
- Consistent reliability during repeated use.

This requires precise measurements, proper mounting brackets, and iterative physical testing.

Expected Outcomes

By the end of the hardware and prototype development stage, the following outcomes are expected:

1. A fully integrated PCB that minimizes wiring complexity and improves reliability.
2. A protective hardware enclosure securely housing all electronics.
3. A complete interactive bus prototype, large enough for a person to enter and test real scenarios.
4. A motorized bus door functions similarly to real school bus doors.
5. Fully installed lights, buzzer, motorized elements, and sensors, connected to the control system.
6. A fully tested system capable of demonstrating:
 - human presence detection.
 - passenger counting.
 - emergency responses.
 - and real-time activation of procedures.
- A realistic demonstration platform (application or dashboard) suitable for showcasing the project's capabilities.

Next step work plan

PCB Design and Fabrication

In the upcoming days, we will begin designing a custom PCB that integrates the Arduino MKR NB 1500, the Raspberry Pi, the cameras, and all sensors into a single organized board. Our goal is to significantly reduce wiring complexity and ensure a clean, stable layout for the entire system. After finalizing the schematic and layout, we plan to fabricate the PCB and carry out continuity, power, and functionality testing to verify that all components communicate correctly.

Building the Component Enclosure

Once the PCB is completed, we will design and fabricate a protective enclosure to house all electronics. This enclosure will include ventilation slots, cable-routing channels, and mounting points to ensure durability and thermal safety. After assembly, we will install this enclosure on the ceiling of the prototype structure, simulating the actual installation location inside a real school bus.

Constructing the Interactive Bus Prototype

During this stage, we will construct a full interactive bus prototype using lightweight MDF wood. This model will include basic interior elements such as a seat, windows, and structural framing to mimic a real bus environment. We will prepare designated mounting areas for each sensor and camera to ensure accurate positioning and reliable testing conditions.

Motorized Bus Door Development

A major focus of this stage will be designing and implementing a functional motorized bus door. We will select suitable materials for the door panel, then test different motor mechanisms such as a high-torque servo motor, a geared DC motor, or a linear actuator, or other possible solutions to determine the most effective solution. After installing the motor, hinges, door frame, and control circuitry, we will perform repeated motion tests to adjust alignment and ensure smooth, stable operation.

Installing Lights, Buzzer, and External Components

We will install lights that mimic real bus lights, as well as a buzzer to simulate the bus horn.

These components will be wired to the Arduino and tested to confirm synchronized activation with the system's detection events and procedures.

Phase-6: Integration and Testing

Once the hardware and interactive prototype are fully assembled, we will move into the Integration and Testing Phase. This stage focuses on connecting all hardware subsystems sensors, cameras, the PCB, the door mechanism, lighting modules, and the Raspberry Pi–Arduino communication layer into one unified operational system. The goal is to validate functionality, uncover hidden issues, verify performance under realistic conditions, and ensure that all system interactions behave as intended. This phase is essential to confirm that the hardware, embedded software, and machine-learning components operate cohesively inside the prototype environment.

Objectives

- Ensure seamless communication between all modules: Arduino, Raspberry Pi, sensors, cameras, and actuators.
- Validate the correct operation of each subsystem after being installed in the prototype.
- Perform end-to-end scenario testing (child detection, door event monitoring, movement identification, alarms triggering).
- Identify performance issues, signal interference, latency problems, or power-related instabilities.
- Refine hardware positioning, cable routing, and mechanical alignment inside the prototype.
- Confirm that the system meets responsiveness, reliability, and safety targets.
- Prepare the system for final demonstration and evaluation.

Potential Challenges

System-Level Communication and Synchronization

- Ensuring real-time communication between the Arduino and Raspberry Pi without data loss or delays.
- Managing multiple data sources (C1001 mmWave, GPS, load sensor, cameras) running simultaneously.
- Avoiding UART/I2C conflicts and ensuring stable data throughput during tests.

Sensors and Cameras Alignment Inside the Prototype

- Achieving optimal camera field of view for child detection.
- Adjusting sensor angles to minimize blind spots or false triggers.
- Ensuring the mmWave sensor's readings are stable when installed near other components.

Power Management

- Managing voltage drops when motors, lights, and buzzer activate simultaneously.
- Ensuring the PCB routes power cleanly to all modules.

Door Mechanism Integration

- Validating that the mechanical door behaves consistently under repeated operation.
- Checking the mechanics and smoothness of the door's movement, and the motor's ability to open and close the door easily and without any problems in the long run.

Full System Scenario Testing

- Simulating realistic behaviors inside the model bus.
- Ensuring cameras detect a child reliably under different lighting conditions.
- Verifying alarms activate correctly and without unintended triggers.
- System debugging and performance evaluation.

Expected Outcomes

By the end of the Integration and Testing Phase, we expect to achieve a fully functional and unified system operating inside our interactive bus prototype. We anticipate that all sensors, cameras, and actuators will be tested and verified under simulated real-world conditions, allowing us to confirm their performance and stability once they are integrated together.

We also expect to establish a robust and reliable communication layer between the microcontrollers and processing units, ensuring smooth coordination between the Arduino, Raspberry Pi, and the various hardware modules.

Throughout the testing process, we will document all results, identify any issues that appear during integration, and apply the necessary adjustments or fixes.

Phase-7: Finalization, Report, and Presentation

The Finalization, Report, and Presentation phase represents the concluding stage of our project, where we focus on adding the final touches, preparing all required documentation, and ensuring the system is ready for demonstration. After months of development, testing, and refinement, this phase allows us to consolidate our work and present the project in a clear, organized, and professional manner. As a team, we will divide tasks strategically to complete all deliverables on time and ensure that every output accurately reflects the effort, experience, and knowledge we gained throughout the project lifecycle.

Objectives

The primary goals of this phase are:

- To finalize the full system and ensure it is stable, functional, and ready for public demonstration.
- To complete all required documentation, including the final report, project poster, and presentation slides.
- To present the project in a comprehensive and visually clear manner, highlighting the technical achievements, challenges, and real-world impact.

- To prepare each team member for the presentation and ensure that roles, content, and narration are well-coordinated.

Potential Challenges

During this closing phase, several challenges may arise:

- **Time Constraints:** Completing documentation, preparing the presentation, and finalizing the prototype must all be done within a limited timeframe.
- **Ensuring Accuracy in Reporting:** Summarizing months of technical work in a concise, clear, and detailed report requires careful coordination and verification.
- **Presentation Readiness:** Preparing a well-organized and confident presentation may require multiple rehearsals to ensure that all team members are aligned.
- **Unexpected Final Adjustments:** Last-minute system tweaks or fixes may arise during the final review or demonstration preparation.

Expected Outcomes

By the end of this phase, we expect to achieve:

- A fully finalized and stable prototype that is ready for demonstration during the project exhibition.
- A complete and accurate final report documenting all phases of the project.
- A professional poster that summarizes the project visually and highlights the key results, contributions, and possible future developments
- A structured and engaging presentation, supported by slides, that clearly communicates the concept, methodology, challenges, changes, and measurable outcomes.

Gantt Chart Modifications

As we progressed through the different stages of the project, our Gantt chart naturally underwent several updates to reflect the actual pace of work and the challenges encountered along the way. Although our initial timeline was planned carefully at the beginning of the semester, a number of unforeseen circumstances required us to adjust the schedule to maintain alignment between tasks and deliverables.

One of the main reasons for modifying the timeline was the delay in receiving some of the critical components, specifically the Arduino MKR NB 1500 and the Arduino Nicla Vision camera due to shipping issues beyond our control. Additionally, the GPS module arrived damaged, which forced us to contact the supplier, return the faulty unit, and wait for a replacement. These issues collectively pushed back the start of several tasks' dependent on hardware availability.

Another significant timeline impact came from the challenges faced with the Arduino Nicla Vision camera. After initially selecting and working on this camera for several days, we discovered critical limitations and performance issues that prevented it from meeting the requirements of our project, especially for live streaming and reliable face-based child counting. As a result, we made a strategic decision to switch to a Raspberry Pi-based solution with two cameras instead. This transition required additional research, configuration, and testing time that was not included in the original Gantt chart, leading to further adjustments.

Furthermore, the technical challenge involving the GPS module and the C1001 mmWave sensor both requiring the same communication interface introduced an unexpected investigation phase. This required us to explore alternative solutions such as I2C-based UART expanders and specialized GPS-to-I2C cables. This exploratory work caused a slight shift in the timeline for software development and early integration tasks.

Despite these changes, updating the Gantt chart allowed us to maintain realistic progress tracking and ensured that the project continued moving forward in an organized manner. The continuous adjustments helped us redistribute workload, adapt to new technical findings, and preserve the overall quality of the project without compromising any major milestone.

Overall, these timeline modifications were a natural and expected part of the development process. They provided us with the flexibility needed to respond to real challenges and positioned us to complete the remaining phases with clearer expectations and a more robust project foundation.

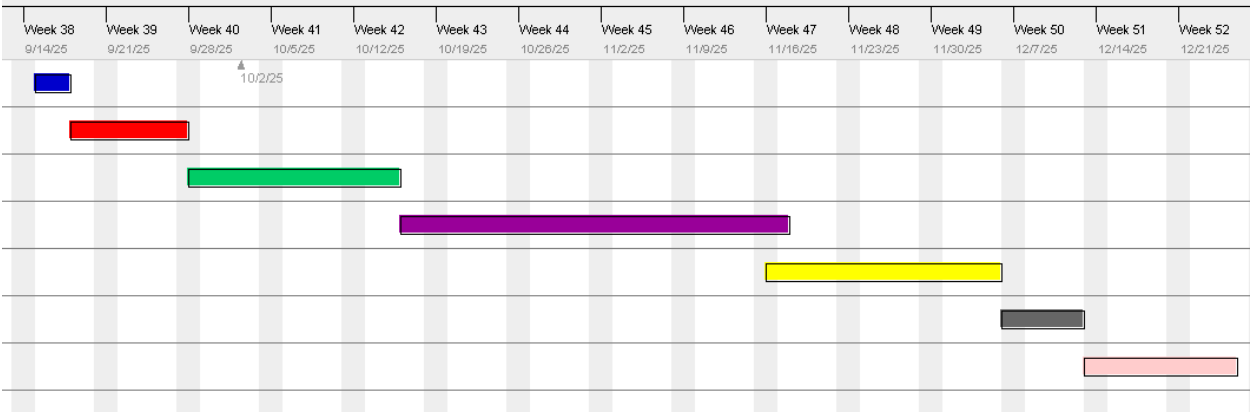


Figure 4: The Modified Gantt Chart

Name		Begin date	End date	Duration
Identifying The Problem	●	9/15/25	9/17/25	3
Research and Planning	●	9/18/25	9/27/25	10
Select the Components	●	9/28/25	10/15/25	18
Software Development	●	10/16/25	11/17/25	33
Hardware Development	●	11/16/25	12/5/25	20
Integration and Testing	●	12/6/25	12/12/25	7
Finalization, Report, and Presentation	●	12/13/25	12/25/25	13

Figure 5: Modified Gantt Chart's Details

Conclusion and Recommendations

Throughout this project, we have navigated a wide range of technical, logistical, and design challenges that gradually shaped our system into a more mature and reliable solution. Our progress so far reflects months of continuous learning, experimentation, and problem-solving. From the early stages of defining the concept, through the difficulties in component sourcing, all the way to debugging complex sensors and redesigning the camera system, every step contributed to strengthening our understanding of both embedded systems and real-world constraints.

Although the project did not always progress exactly as planned, the obstacles we faced ultimately pushed us to develop more robust solutions. The shift from the Arduino Nicla Vision camera to the Raspberry Pi system, the struggle with the C1001 mmWave calibration, the sensor-interference issues, and the GPS communication conflict each played a role in improving our technical decisions and preparing a more dependable final product. These experiences also helped us refine our teamwork, adaptability, and research skills elements that have become just as valuable as the technical outcomes themselves.

Recommendations

Based on the challenges we encountered so far, we believe several recommendations can benefit future project cycles:

1. Begin Hardware Procurement Much Earlier

Many delays were caused by late shipments, damaged components, or unavailable parts. Securing all hardware early would have prevented several timeline shifts and allowed us to focus more on development rather than waiting for deliveries.

2. Validate Component Capabilities Before Integration

Our experience with the Nicla Vision camera showed the importance of performing early capability checks. Testing hardware limitations early, especially for critical features such as live-stream video or face detection can save significant redesign time.

3. Allocate Extra Time for Sensor Calibration and Debugging

Sensors such as the C1001 mmWave, load cell (HX711), and IR alternatives required far more time than initially expected. Future teams should allocate larger buffer periods for calibration, environmental testing, and fine-tuning.

4. Maintain a Flexible and Continuously Updated Gantt Chart

Because real-world development rarely follows a perfect timeline, the Gantt chart should be treated as a dynamic tool. Regular updates helped us adapt to new findings without losing track of overall progress.

5. Document Every Stage Thoroughly

The detailed records we kept during testing, debugging, and decision-making became extremely valuable when writing this report and preparing for the next stages. Continuous documentation helps track insights, mistakes, and lessons learned.

6. Consider Redundancy in System Design

Introducing backup sensors (such as adding a thermal sensor to support presence detection) significantly enhances reliability. Building redundancy into the design from the start can reduce false alarms and improve system performance.